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|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--|
| SUMMARY             | I study how large language models reason under uncertainty, with a focus on adaptive computation and efficient architectures for complex reasoning tasks.                                                                                                                                                                                                           |                                          |  |
| EDUCATION           | University of Southern California                                                                                                                                                                                                                                                                                                                                   | Los Angeles, CA                          |  |
|                     | Ph.D., Master's in Computer Science                                                                                                                                                                                                                                                                                                                                 | August 2025 - Expected 2029, 2023 - 2025 |  |
|                     | <ul style="list-style-type: none"><li>Adaptive Model Reasoning, Long-context Models, Reinforcement Learning</li><li>Natural Language Processing, Computer Vision, Algorithms, OS</li></ul>                                                                                                                                                                          |                                          |  |
|                     | Stanford University                                                                                                                                                                                                                                                                                                                                                 | Stanford, CA                             |  |
|                     | Master's, Bachelor's in Mechanical Engineering                                                                                                                                                                                                                                                                                                                      | 2013 - 2019                              |  |
|                     | <ul style="list-style-type: none"><li>Manufacturing, Robotics, Programming, Applied Machine Learning</li></ul>                                                                                                                                                                                                                                                      |                                          |  |
| PUBLICATIONS        | 1. Ryan Lee, Jacob Biloki, Edward J. Hu, Jonathan May. "Sparse Layers are Critical to Scaling Looped Language Models." <i>Advances in Neural Information Processing Systems (NeurIPS)</i> , 2026.                                                                                                                                                                   |                                          |  |
|                     | 2. Xuezhe Ma*, Shicheng Wen*, Linghao Jin*, Bilge Acun*, Ruihang Lai*, Bohan Hou, Will Lin, Hao Zhang, Songlin Yang, Ryan Lee, Mengxi Wu, Jonathan May, Luke Zettlemoyer, Carole-Jean Wu. "Gecko: An Efficient Neural Architecture Inherently Processing Sequences with Arbitrary Lengths." <i>Third Conference on Language Modeling (COLM)</i> , October 2026.     |                                          |  |
| RESEARCH EXPERIENCE | Adaptive-Depth Language Models   USC ISI                                                                                                                                                                                                                                                                                                                            | May 2025 - May 2026                      |  |
|                     | <ul style="list-style-type: none"><li>Found sparse MoE layers resolve the scaling deficit of looped transformers.</li><li>Looped-MoE beat dense LMs on OLMES (39.6 vs 38.7) with 12% fewer params.</li><li>Looped models degrade less on early exit: 51.0 vs 55.4 ppl at 10% FLOPs saved.</li></ul>                                                                 |                                          |  |
|                     | LLM Pretraining & Long Context Legal Retrieval   USC ISI                                                                                                                                                                                                                                                                                                            | Nov 2024 - May 2025                      |  |
|                     | <ul style="list-style-type: none"><li>Pretrained Gecko 1B on a 1.7T token corpus, outperforming OLMo1.</li><li>Developed IR pipelines using FAISS for efficient dense retrieval and ranking.</li></ul>                                                                                                                                                              |                                          |  |
| INDUSTRY EXPERIENCE | Netflix, Inc.   Machine Learning Engineer Intern   Los Gatos, CA                                                                                                                                                                                                                                                                                                    | May 2026 - Aug 2026                      |  |
|                     | <ul style="list-style-type: none"><li>Rewrote &amp; pretrained a legacy model with GQA, RMSNorm, and RoPE for personalized generation of rows of titles on the Netflix homepage.</li><li>Developed cold-start recommendation with semantic IDs (cold recall 0→0.113).</li><li>Post-trained an LLM for homepage row title recommendation via SFT and GRPO.</li></ul> |                                          |  |
|                     | Apple Inc.   Product Designer   Cupertino, CA                                                                                                                                                                                                                                                                                                                       | May 2021 - June 2023                     |  |
|                     | <ul style="list-style-type: none"><li>Led Touch ID development on the M2 MacBook Air through mass production.</li><li>Guaranteed Touch ID by design across &gt;1M units via Monte Carlo and 3-D FEA.</li></ul>                                                                                                                                                      |                                          |  |
|                     | Microsoft Corp.   Mechanical Design   Redmond, WA                                                                                                                                                                                                                                                                                                                   | Sep 2019 - Apr 2021                      |  |
|                     | <ul style="list-style-type: none"><li>Built Surface Slim Pen 2 haptic waveforms via bandpass filtering and FFTs.</li><li>Designed an automated computer-vision pen tester accurate to 500 microns.</li></ul>                                                                                                                                                        |                                          |  |
| SKILLS              | Programming: Python, C/C++, R, Java, Bash/Shell scripting                                                                                                                                                                                                                                                                                                           |                                          |  |
|                     | ML & NLP: PyTorch, HF (Transformers, Datasets), scikit-learn, spaCy, Pandas, NumPy                                                                                                                                                                                                                                                                                  |                                          |  |
|                     | Retrieval: FAISS, LlamaIndex, dense retrieval, RAG systems, embedding models                                                                                                                                                                                                                                                                                        |                                          |  |
|                     | Infrastructure: SLURM, AWS/GCP, Git, Conda/pip, Jupyter, Wandb                                                                                                                                                                                                                                                                                                      |                                          |  |